

From Algorithmic Fairness to Evidentiary Reliability: Judicial Analytics under Article 6 ECHR

ARJUN BHUBANESHWAR BANERJEE, Vrije Universiteit Brussels, Belgium

Data-driven analyses of judicial decision-making now range from traditional empirical legal studies based on inferential statistics to computational systems that deploy technologies like machine learning (ML) and natural language processing (NLP) to detect and predict patterns in judicial decisions. In debates relating to algorithmic fairness, these methods are usually evaluated through bias metrics, mitigation techniques, and predictive performance. This paper asks a normative legal question: under what conditions can such outputs function as legally reliable evidence in the assessment of judicial impartiality under Article 6(1) European Convention on Human Rights (ECHR)? The paper departs from the argument that statistical validity and legal reliability are distinct standards. Statistical validity concerns model performance, robustness, and predictive success. Legal reliability, on the other hand, concerns whether an empirical claim can operate as a reason within adjudication. Situated at the level of the European Court of Human Rights (ECtHR), the paper reconstructs reliability not as a domestic admissibility rule but as a condition of epistemic justification. It shows that empirical or computational claims about judicial decision-making can inform the Article 6 inquiry only if they are grounded in ascertainable facts, intelligible to legal reasoning, open to adversarial scrutiny, and are methodologically credible. The paper develops this claim through a four-part framework of ascertainability, transparency and source accountability, methodological credibility, and explainability [[1], [2], [3], [4], [5]].

Keywords: Algorithmic fairness, judicial analytics, Article 6 ECHR, legal reliability, explainability

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1 Introduction

The use of data-driven methods to analyze judicial decision-making has expanded rapidly. What began in empirical legal studies as inferential modelling of relatively limited, hand-curated datasets increasingly includes computational systems that learn patterns from very large collections of judgments. These approaches should not be collapsed into a single methodological category. Inferential statistical studies ordinarily seek to explain associations between variables through specifications that can be interrogated, revised, and replicated. Predictive systems in machine learning, by contrast, prioritize classification or forecasting performance and may generate outputs whose inferential pathway is only partially accessible to legal users [[6], [7], [8], [9], [10]].

Within contemporary debates on algorithmic fairness, both forms of analysis are typically assessed through the language of statistical disparity, bias mitigation, and model accuracy. That literature has generated important insights, especially on discrimination, data quality, and the opacity of high-performing models. Yet once such outputs are invoked to support

Arjun Bhubaneshwar Banerjee, Vrije Universiteit Brussel, Belgium, arjun.bhubaneshwar.banerjee@vub.be

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claims about judicial bias, a different question becomes unavoidable: when do they become legally reliable evidence? This question becomes critical in relation to judicial analytics, where empirical or computational analyses are increasingly used to support propositions about systemic tendencies, institutional risk, or the appearance of partiality in adjudication [[6], [7], [8], [9]].

2 The Reliability problem in Article 6 Adjudication:

This paper addresses that question in the specific context of Article 6(1) ECHR. Its claim is not that quantitative evidence should be excluded from impartiality adjudication. On the contrary, the Strasbourg case law is already open to appearance-based reasoning and indirect proof. In *Piersack and De Cubber*, the ECtHR made clear that the decisive issue under the objective test of judicial impartiality is whether doubts about impartiality can be regarded as objectively justified, on the basis of ascertainable facts rather than subjective suspicion [[1], [2]]. In *El-Masri*, the Grand Chamber further held that proof may follow from “sufficiently strong, clear and concordant” inferences, thereby confirming that Convention adjudication does not depend exclusively on direct evidence [[3]]. And in its Article 14 jurisprudence, especially *D.H. and Biao*, the Court has accepted that statistical patterns may carry probative force when they are reliable and significant enough to establish a *prima facie* claim [[4], [5]].

The doctrinal problem, therefore, is not whether empirical material is admissible in principle. It is what makes it reliable enough to carry probative weight under Article 6. The paper argues that a fundamental distinction must be drawn between statistical validity and legal reliability. Statistical validity is usually expressed in terms such as predictive accuracy, robustness, or out-of-sample performance. Whereas the core inquiry under legal reliability is different. It asks whether an empirical claim can function as a legally usable reason: whether it can be traced to verifiable material, translated into an intelligible inferential claim, contested by the parties, and used by a court without overstating what the evidence can bear. That distinction matters for judicial analytics.

Early work on ECtHR prediction and later large-scale judicial forecasting studies show that legal outcomes can be modelled with non-trivial success [[6], [7]]. But subsequent research also demonstrates that such systems may rely on shallow or legally irrelevant signals, and that deconfounding and explanation are central if predictive outputs are to align with expert legal judgment [[8], [9]]. As Epstein and King argued in the broader context of empirical legal scholarship, methodological force does not remove the need for disciplined inference [[10]].

The central research question is therefore:

Under what conditions can statistical and machine-learning analyses of judicial decision-making be regarded as legally reliable evidence for the purposes of Article 6 ECHR?

3 Argument and contribution:

The paper situates this inquiry at the level of the ECtHR rather than within any single domestic evidentiary regime. The point is not to compare national admissibility rules, which vary significantly across legal systems, but to reconstruct the pan-European normative conditions under which empirical or computational outputs may justify conclusions about judicial impartiality under Article 6(1). This level of analysis is especially apt because current European governance

instruments already identify justice as a sensitive site for AI regulation. The CEPEJ Ethical Charter treats transparency, fairness, and user control as central concerns in the use of AI in judicial systems [[11]]. Likewise, the EU AI Act classifies AI systems used by or on behalf of judicial authorities to assist in researching and interpreting facts and law, and in applying the law to a concrete set of facts, as high-risk systems [[12], Annex III, point 8(a)]. Such systems are subject to requirements relating to risk management, technical documentation, traceability, transparency, human oversight, accuracy, robustness, and cybersecurity [[12], arts. 9–15]. These instruments do not determine Article 6 doctrine. But they do confirm that, in the European legal order, reliability in justice-adjacent AI cannot be reduced to predictive performance alone.

Against that background, the paper proceeds in three steps. First, it reconstructs reliability within Article 6 jurisprudence by focusing on the Court's treatment of objective impartiality, inferential proof, and evidentiary substantiation [[1], [2], [3], [4], [5]]. Second, it distinguishes inferential empirical models from predictive computational systems and shows why that epistemic difference matters for legal use [[6], [7], [8], [9], [10]]. Third, it develops a four-part framework for legal reliability: ascertainability, transparency and source accountability, methodological credibility, and explainability.

Ascertainability concerns whether the evidentiary basis of an empirical claim can be traced to verifiable facts. Transparency and source accountability require disclosure of dataset provenance, coding decisions, and model design such that opposing parties and the court can scrutinize the inferential path. Methodological credibility asks whether the observed association is robust enough to survive obvious rival explanations, including confounding, selection effects, or unstable proxy variables [[8], [10]]. Explainability addresses a further challenge that is critical for machine-learning systems: whether a predictive output can be translated into reasons that are legally intelligible rather than offered as a technically impressive but opaque score. This last requirement is particularly important in the context of adjudication, where contestability and reason-giving are not optional virtues but structural features of legal legitimacy [[9], [11], [12]].

The contribution of the paper is therefore twofold. Substantively, it reframes judicial analytics not as a problem of abstract model fairness alone, but as a problem of evidentiary translation within the law on fair trial. Conceptually, it shows why fairness metrics and predictive success are insufficient unless they are converted into a form of knowledge capable of grounding objective, contestable, and reasoned adjudication under Article 6. The broader implication is that European debates on fairness and transparency need a more explicit account of legal reliability when algorithmic tools enter judicial and quasi-judicial settings. By shifting the focus from whether algorithmic systems are statistically fair to whether their outputs can satisfy the evidentiary demands of adjudication, the paper offers a framework that speaks directly to current concerns about black-box models, AI in the judiciary, and the normative limits of data-driven decision support in Europe [[8], [9], [11], [12]].

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